Scalor, vector. Bazil-uddin-khan

Vector:- **Scalar:-** → only magnitude/unit can be destribed by it.

Ex:- 5kg. , volume, length, mass etc.

Vector:- Magnitude, direction both , e.g:- displacement etc.

Vector operations → Resolution into components and reverse calculation of Polar form? which are used in → Graphical method → Addition of vectors

→ Scalor Product → can be multiplied with a → in work calculation etc.

→ Vector Product → can be m.. with a → Torque, magnetic force.

Simplest vector is displacement vector → or change in position.

Adding vectors geometrically:-

"Resultant vector" is two or more vector that represent a combined vector"

→ Properties of vector addition:- $\vec{AS} = \vec{a} + \vec{b}$

1-commutative → $\vec{a} + \vec{b} = \vec{b} + \vec{a}$

2-Associative → $(\vec{a} + \vec{b}) + c = a + (b+c)$.

Vector subtraction:- $\vec{b}$ $-\vec{b}$ → $A - b = \boxed{A + (-b)}$ look like addition just subtraction.

components of vectors:- $a_x = a\cos\theta$, $a_y = a\sin\theta$, components, vector form a right angle.

$a = \sqrt{a_x^2 + a_y^2}$ and $\tan\theta = \left|\frac{a_y}{a_x}\right|$ (angle measured through +ve axis always)

In 3rd Quadrant → $180^\circ + \theta$

θ) → $|a| = |b|$

i) $|a+b| = 0$ (b) $|a+b| = 2|A|$, (c) $|a+b| = \sqrt{2}a$

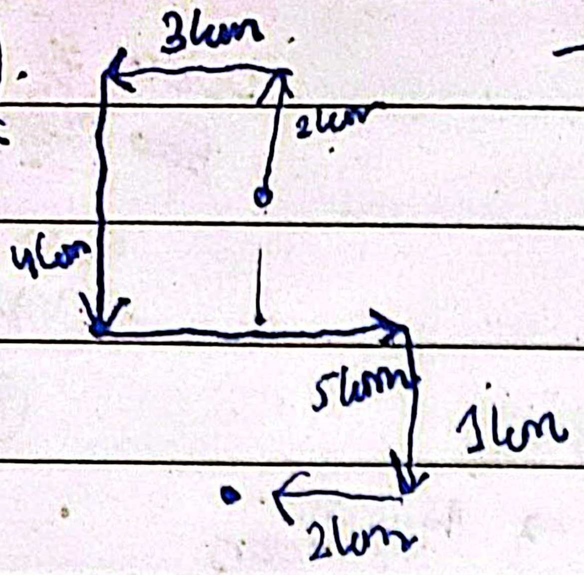
→ $0 = a^2 + a^2 + 2a^2\cos\theta$ ↳ $4a^2 = a^2 + a^2 + 2ab\cos\theta$ ↳ $\sqrt{2}a^2 = a^2 + a^2 + 2a^2\cos\theta$

$-2a^2 = 2a^2\cos\theta$ $0 = \cancel{2ab\cos\theta}$ $2a^2 = 2a^2\cos$ $0 = 2a^2 = 2a^2 + 2a^2\cos\theta$

$\theta = 180$ $\theta = 90^\circ$ $\theta = 0^\circ$ $\theta = 90^\circ$

Q).

X



$\rightarrow 17,$

Q) $A = 3i + 7j - 8k$

Ans: $\dfrac{3i + 7j - 8k}{\sqrt{9+49+64}} \rightarrow \boxed{\dfrac{3i+7j-8k}{\sqrt{122}}}$

$$\begin{array}{r} 249 \\ 64 \\ \hline 122 \end{array}$$

$d_1 + (-d_2)$

Q). $(4.2 + 1.6 + 0\,i) + (-1.5 + 2.9 - 3.7)$

$$\begin{array}{r} 2.7 \\ 1.5 \\ \hline 5.2 \\ 2.9 \\ \hline \end{array}$$

Ans:- $2.6i - 2.3j$

$\tan\theta = -\dfrac{2.3}{2.6} \rightarrow \dfrac{\frac{23}{10}}{\frac{26}{10}} \rightarrow \theta = \tan^{-1}\left(\left|\pm\dfrac{23}{26}\right|\right)$

$\boxed{\theta = 41.5^\circ} \rightarrow 360 - 41.5 = 318.6$

$R = \sqrt{(2.6)^2 + (2.3)^2} = \boxed{R = 3.5}$